Off-Grid SBL Channel Estimation With Path-Support Learning for AFDM

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***Abstract*—Affine frequency division multiplexing (AFDM) is a promising multicarrier waveform for sixth-generation (6G) high-mobility communications, whose performance depends on accurate channel estimation. However, mismatch between continuous Doppler shifts and a predefined grid introduces modeling errors, while high correlation among candidate atoms makes path identification difficult. Noise at low signal-to-noise ratios further aggravates this difficulty. To address these issues, this paper proposes a sparse Bayesian learning method with a Bernoulli–Gaussian prior (BG-SBL) that combines path-support learning with continuous-Doppler dictionary refinement and uses expectation/conditional maximization either (ECME) for inference. In addition, a known-support Cramér–Rao lower bound (CRLB) is derived to characterize the theoretical estimation limit. Simulation results show that the proposed BG-SBL method reduces channel-estimation normalized mean square error, suppresses redundant and spurious path estimates, and improves path identification and continuous-Doppler estimation.**

***Index Terms*—Affine frequency division multiplexing (AFDM), path-support learning, off-grid channel estimation, sparse Bayesian learning (SBL), marginal evidence, Cramér–Rao bound (CRB).**

I. Introduction

HIGH-mobility connectivity is expected to become a fundamental capability of sixth-generation (6G) wireless systems, with representative applications including high-speed railway communications, vehicular networks, unmanned aerial vehicles, and low-Earth-orbit satellite communications. In such scenarios, rapid temporal channel variations result in pronounced Doppler shifts. Although orthogonal frequency division multiplexing (OFDM) is highly effective over slowly varying frequency-selective channels, Doppler shifts destroy subcarrier orthogonality and introduce severe inter-carrier interference (ICI), leading to substantial performance degradation [1]. This limitation has motivated the development of alternative waveforms that can better exploit the intrinsic structure of doubly dispersive channels.

Orthogonal chirp division multiplexing (OCDM) [2] improves Doppler robustness by transmitting information over orthogonal chirp waveforms. However, its fixed chirp-rate configuration limits its adaptability and attainable performance in highly dynamic propagation environments. Orthogonal time frequency space (OTFS) modulation, on the other hand, maps information symbols onto the delay–Doppler domain and can effectively exploit the underlying delay–Doppler diversity of the channel, thereby providing significant performance advantages over OFDM in doubly dispersive environments [3]. Nevertheless, the two-dimensional transformations between the time–frequency and delay–Doppler domains introduce additional computational complexity [5]. AFDM provides an alternative chirp-based solution with two adjustable chirp parameters that can be designed according to the delay and Doppler spreads of the channel [4]. With appropriate parameter selection, AFDM yields a structured delay–Doppler representation in the discrete affine Fourier transform (DAFT) domain and can achieve full diversity over doubly dispersive channels [4], [5]. Compared with the two-dimensional transforms used in OTFS, AFDM employs a one-dimensional DAFT, simplifying the transform structure and reducing modulation and demodulation complexity [5]. These properties make AFDM particularly attractive for high-mobility communications. Accurate channel estimation is essential to provide reliable channel state information (CSI) for receiver equalization and realize AFDM's diversity gains.

Existing AFDM channel estimation methods can be categorized into non-SBL methods and those based on Sparse Bayesian Learning (SBL). Non-SBL methods mainly include OMP and diagonal reconstruction. Cao et al. applied OMP to AFDM transmission over multi-scale multi-lag channels with Doppler time-scaling effects and demonstrated its effectiveness in the corresponding propagation scenarios [9]. However, the use of a discrete dictionary fundamentally limits the attainable parameter resolution. Ma et al. further proposed a Newtonized channel-estimation method that recovers fractional delays and Doppler shifts through coarse estimation, single-path refinement, and cyclic refinement, while precomputation is employed to reduce the computational cost associated with response vectors and their derivatives [10], [12]. Nevertheless, such methods remain inherently dependent on greedy sequential estimation, and the cyclic refinement steps introduce non-negligible computational overhead. For MIMO-AFDM systems, Yin et al. proposed a diagonal reconstruction approach that exploits the reconstructability of the diagonals of the subchannel matrices to recover the equivalent channel without explicitly estimating the delay, Doppler, and gain of each individual propagation path [11]. This strategy reduces computational complexity and alleviates inter-Doppler interference encountered in path-wise estimation. However, it primarily reconstructs the equivalent channel matrix required for receiver equalization and does not directly recover the underlying continuous path parameters, thereby limiting its achievable parameter-estimation accuracy.

Compared with the aforementioned non-SBL methods, SBL employs a hierarchical sparse prior to achieve adaptive sparse recovery, demonstrating robust sparse reconstruction capabil-

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ities in the presence of noisy observations [13], [14]. Grid-based SBL constructs a dictionary composed of pilot response vectors (referred to as dictionary atoms) for candidate delay-Doppler pairs on a predefined grid and estimates their sparse coefficients [20]. However, practical Doppler shifts are continuous and generally do not coincide with the grid points, resulting in grid mismatch and degraded estimation accuracy [22]. To alleviate this problem, off-grid SBL (OG-SBL) employs a first-order Taylor expansion to approximate off-grid components and improves estimation accuracy [21], [22]. Nevertheless, its underlying grid remains fixed, resulting in a noticeable error floor at high signal-to-noise ratios (SNRs). Grid-refinement SBL (GR-SBL) and grid-evolution SBL (GE-SBL) reduce grid mismatch through dynamic Doppler-grid updates. GR-SBL achieves high parameter resolution through local searches at increased computational cost as the search step size decreases, whereas GE-SBL uses first-order approximations to evolve the grid and offers a more favorable accuracy–complexity tradeoff [23]. Although the aforementioned SBL methods mitigate grid mismatch through off-grid modeling or grid refinement, the dictionary may still contain candidate atoms with identical delays and similar Doppler shifts. These candidates correspond to highly similar dictionary atoms—and thus exhibit high correlation—potentially resulting in a single physical path being jointly represented by multiple adjacent active components. When a fixed threshold is applied to detect these components individually, any components representing the same path that exceed the threshold are separately identified as valid paths; this leads to an overestimation of the number of paths and degrades channel estimation performance.

To mitigate the issues arising from Doppler grid mismatch and high dictionary correlation, we propose a Bernoulli–Gaussian Sparse Bayesian Learning (BG-SBL) algorithm for off-grid channel estimation in AFDM systems. This algorithm integrates path support learning with continuous Doppler refinement. The main contributions are summarized as follows:

- First, the off-grid error is modeled as a Doppler-induced dictionary perturbation. Subsequently, AFDM channel estimation under dictionary mismatch is formulated as a joint optimization problem involving parametric dictionary learning and sparse channel recovery, which is solved within the SBL framework.
- We have developed a BG-SBL algorithm that explicitly models the active status of candidate paths using Bernoulli variables. The algorithm iteratively adds, retains, and removes paths based on a unified criterion for path existence, while simultaneously updating the Doppler coordinates of true paths using Newton's method.
- A known-support Cramér–Rao bound is derived under a moment-matched colored-Gaussian likelihood for the parameterized AFDM model, with the complex path gains, continuous Doppler offsets, and thermal-noise variance treated as unknown. The proposed estimator is comprehensively evaluated against representative AFDM baselines in terms of channel-estimation NMSE, path-detection accuracy, Doppler estimation accuracy, and bit-error rate.

II. AFDM System Model

This section first presents the input–output relationship of an AFDM system over a doubly dispersive channel and then derives the pilot observation model used for channel estimation.

A. AFDM Transmission Over Doubly Dispersive Channels

Following [4], consider an N-subcarrier AFDM block with subcarrier spacing Δf and sampling interval $T_s = 1/(N\Delta f)$. Let $\boldsymbol{F}$ be the unitary DFT matrix, with $[\boldsymbol{F}]_{m,n} = N^{-1/2}e^{-j2\pi mn/N}$, and define the chirp vector $\boldsymbol{\omega}_c = [1, e^{-j2\pi c}, \dots, e^{-j2\pi c(N-1)^2}]^T$. Then,

$$\boldsymbol{\Lambda}_c = \mathrm{diag}(\boldsymbol{\omega}_c), \qquad \boldsymbol{A} = \boldsymbol{\Lambda}_{c_2}\boldsymbol{F}\boldsymbol{\Lambda}_{c_1}. \tag{1}$$

The DAFT matrix $\boldsymbol{A}$ is unitary, and the transmitted time-domain block is $\boldsymbol{x} = \boldsymbol{A}^H\boldsymbol{s}$ for a DAFT-domain symbol vector $\boldsymbol{s} \in \mathbb{C}^N$. A chirp-periodic prefix of length at least $\ell_{\max}$ suppresses interblock interference, where $\ell_{\max}$ denotes the maximum allowed integer delay in sampling intervals. We assume that N is even and $2Nc_1$ is an integer, for which the prefix reduces to a conventional cyclic prefix and no additional prefix-phase matrix is required.

For a doubly dispersive channel with P paths, the time-varying impulse response is

$$h[n,\ell] = \sum_{p=1}^{P} h_p e^{-j2\pi\nu_p n/N}\delta[\ell-\ell_p], \qquad \nu_p = k_p + \beta_p, \tag{2}$$

where $h_p \in \mathbb{C}$, $\ell_p \in \mathcal{L} \triangleq \{0,\dots,\ell_{\max}\}$, and ν_p denote the complex gain, integer delay, and continuous normalized Doppler of the pth path, respectively. Here, $p = 1,\dots,P$ indexes the paths, while ℓ_p specifies the delay in sampling intervals and may be zero. The physical Doppler shift is $f_{D,p} = \nu_p\Delta f$. The normalized Doppler is decomposed into $|k_p| \le k_{\max}$ and $\beta_p \in (-1/2, 1/2]$. We assume that the system bandwidth provides sufficient delay resolution to neglect fractional-delay effects [23]. Accordingly, path delays are modeled as integer multiples of T_s, while Doppler shifts remain continuous.

Let $\boldsymbol{\Pi}$ denote the cyclic-shift matrix, $[\boldsymbol{\Pi}^\ell\boldsymbol{x}]_n = x([n-\ell]_N)$. Define the Doppler phase vector and its associated diagonal matrix as $\boldsymbol{d}(\nu) = \left[e^{-j2\pi\nu n/N}\right]_{n=0}^{N-1}$ and $\boldsymbol{\Delta}(\nu) = \mathrm{diag}(\boldsymbol{d}(\nu))$. After prefix removal and DAFT demodulation,

$$\boldsymbol{r} = \sum_{p=1}^{P} h_p\boldsymbol{\Delta}(\nu_p)\boldsymbol{\Pi}^{\ell_p}\boldsymbol{x} + \widetilde{\boldsymbol{n}}, \tag{3a}$$

$$\boldsymbol{y}_{f\omega} = \boldsymbol{A}\boldsymbol{r} = \boldsymbol{H}\boldsymbol{s} + \boldsymbol{n}_{f\omega}, \tag{3b}$$

$$\boldsymbol{H} = \sum_{p=1}^{P} h_p\boldsymbol{A}\boldsymbol{\Delta}(\nu_p)\boldsymbol{\Pi}^{\ell_p}\boldsymbol{A}^H, \tag{3c}$$

where $\boldsymbol{y}_{f\omega}$ denotes the received signal vector in the DAFT domain and $\boldsymbol{n}_{f\omega} = \boldsymbol{A}\widetilde{\boldsymbol{n}}$ is the corresponding noise vector. Unitarity of $\boldsymbol{A}$ preserves white Gaussian noise statistics.

The chirp parameters are known waveform settings. We choose c_2 to be an irrational number. A sufficient choice for separating delay–Doppler signatures is [4], [23]

$$c_1 = \frac{2k_{\max} + 2N_v + 1}{2N}, \\ N > (2k_{\max} + 2N_v + 1)(\ell_{\max} + 1), \tag{4}$$

where $N_v \geq 0$ is the integer guard margin used in the AFDM parameter and pilot-guard design.

B. Embedded Pilot Structure and Observation Model

To separate the responses of pilot and data signals, we employ contiguous pilot blocks and zero guard zones to construct an embedded pilot structure. Let N_p denote the number of consecutive pilots. Their zero-based index set is

$$\mathcal{P} = \{m_0, m_0+1, \ldots, m_0+N_p-1\}, \quad m_0 = N/2-1, \tag{5}$$

and the guard length on either side is $G = (\ell_{\max}+1)(2k_{\max}+2N_v+1)-1$. The known pilot vector $\boldsymbol{s}_p$ is supported on $\mathcal{P}$, the data vector $\boldsymbol{s}_d$ occupies the remaining unguarded entries, and $\boldsymbol{s} = \boldsymbol{s}_p + \boldsymbol{s}_d$.

Substituting the DAFT entries into the path response in (3c) gives the delay–Doppler-dependent factor

$$F_p(\widetilde{m}, m) = \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}(\widetilde{m}-m+2Nc_1\ell_p+\nu_p)n}, \tag{6}$$

where m and $\widetilde{m}$ denote the transmit and receive DAFT indices, respectively. For integer Doppler ($\beta_p = 0$), this factor is nonzero only at $\widetilde{m} = m - 2Nc_1\ell_p - k_p$ modulo N. Fractional Doppler spreads the response around this index, and the margin N_v specifies the neighboring receive indices retained for estimation.

Taking $m \in \mathcal{P}$, $0 \leq \ell_p \leq \ell_{\max}$, and $|k_p| \leq k_{\max}$, the retained receive window extends from the response of the first pilot under the largest delay and positive Doppler to that of the last pilot under zero delay and negative Doppler. Its endpoints are therefore

$$\widetilde{m}_{\min} = m_0 - 2Nc_1\ell_{\max} - k_{\max} - N_v, \\ \widetilde{m}_{\max} = m_0 + N_p - 1 + k_{\max} + N_v. \tag{7}$$

Using (4) and the definition of G gives $2Nc_1\ell_{\max} = G - 2k_{\max} - 2N_v$, so the lower endpoint becomes $\widetilde{m}_{\min} = m_0 - G + k_{\max} + N_v$. The number of retained samples is $M = \widetilde{m}_{\max} - \widetilde{m}_{\min} + 1 = G + N_p$. We thus define the truncated received pilot signal as

$$\boldsymbol{y}_T = \{y_{f\omega}[\widetilde{m}] \mid \widetilde{m} = \widetilde{m}_{\min}, \ldots, \widetilde{m}_{\max}\}, \tag{8}$$

with entries ordered by increasing receive index. The subscript T denotes truncation. Figure 1 illustrates the embedded pilot structure and the receive window selected from the DAFT-domain output.

The input–output relationship of the truncated received pilot signal is

$$\boldsymbol{y}_T = \sum_{p=1}^{P} h_p \boldsymbol{\phi}(\ell_p, \nu_p) + \boldsymbol{n}, \tag{9}$$

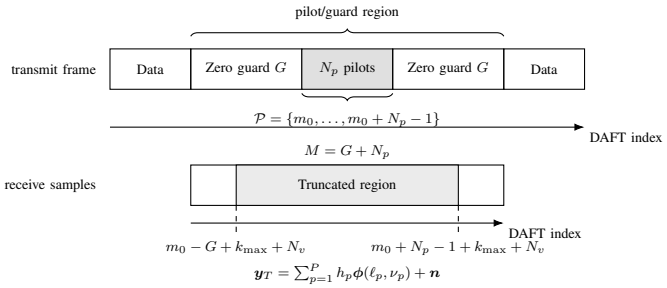


Fig. 1. Embedded pilot structure and selected observation window.

where $\boldsymbol{\phi}(\ell, \nu) = [\phi_0(\ell, \nu), \ldots, \phi_{M-1}(\ell, \nu)]^T \in \mathbb{C}^M$ is the unit-gain pilot-response vector. Its $\bar{m}$th entry is given by

$$\phi_{\bar{m}}(\ell, \nu) = \frac{1}{N} \sum_{m \in \mathcal{P}} s_p[m] e^{j2\pi[c_1\ell^2 - m\ell/N + c_2(m^2 - \widetilde{m}^2)]} \\ \times \frac{e^{-j2\pi(\widetilde{m}-m+2Nc_1\ell+\nu)} - 1}{e^{-j\frac{2\pi}{N}(\widetilde{m}-m+2Nc_1\ell+\nu)} - 1}, \tag{10}$$

with $\widetilde{m} = \widetilde{m}_{\min} + \bar{m}$ and $\bar{m} = 0, 1, \ldots, M-1$. The ratio in (10) is interpreted by its continuous limit when the denominator vanishes. The effective disturbance $\boldsymbol{n}$ includes the thermal noise and any residual pilot–data interference within the truncated receive window, and is modeled as $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \gamma^{-1}\boldsymbol{I}_M)$ for Bayesian inference. The simulations generate the complete pilot-and-data frame, so the residual data interference is not provided separately to the estimator.

III. Parametric Dictionary and Off-Grid Channel Representation

This section uses a fixed Doppler grid only as an initialization and as a baseline for exposing two consequences of grid mismatch: structured modeling error and highly correlated neighboring atoms. We then introduce a parametric dictionary with continuously adjustable Doppler coordinates to represent off-grid AFDM channels.

A. Off-Grid Mismatch

Because only a small number of propagation paths contribute to the received signal, the continuous delay–Doppler channel in (9) can be initialized using a finite set of candidate delay–Doppler coordinates [22], [29]. As discussed in Section II, the path delays are assumed to be integer-valued. In contrast, the normalized Doppler shift is continuous. To construct a reference sparse representation, the Doppler search interval $[-\nu_{\max}, \nu_{\max}]$ is uniformly sampled using

$$r_\nu = \frac{2\nu_{\max}}{M_\nu - 1}, \\ \nu_b = -\nu_{\max} + br_\nu, \qquad b = 0, 1, \ldots, M_\nu - 1.$$

where M_ν denotes the number of reference Doppler-grid points and r_ν is the corresponding grid spacing.

In practical high-mobility channels, however, the physical Doppler ν_p is continuous and generally does not coincide with a predefined Doppler-grid point. Let ν_{b_p} denote the nearest

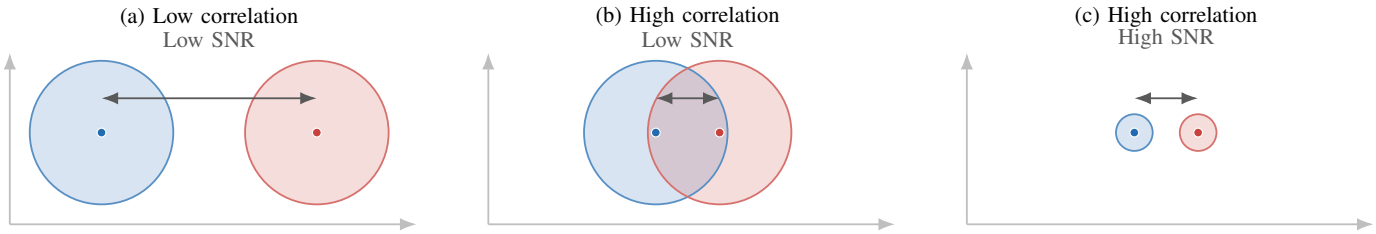


Fig. 2. Distinguishability of atom responses (blue dots: atom 1; red dots: atom 2).

reference Doppler coordinate to the pth physical path and define its Doppler displacement as

$$\varepsilon_p = \nu_p - \nu_{b_p}, \qquad |\varepsilon_p| \le \frac{r_\nu}{2}.$$

When $\varepsilon_p \neq 0$, the exact response $\boldsymbol{\phi}(\ell_p, \nu_p)$ associated with the physical path cannot be accurately represented by the nearest-grid atom $\boldsymbol{\phi}(\ell_p, \nu_{b_p})$. Using this atom therefore introduces a deterministic dictionary mismatch. Importantly, this mismatch is not equivalent to a scalar perturbation of the path gain. The Doppler parameter enters the diagonal phase matrix $\boldsymbol{\Delta}(\nu)$ and consequently changes the phase profile across the observation samples; hence, in general, no single complex coefficient can transform $\boldsymbol{\phi}(\ell_p, \nu_{b_p})$ into $\boldsymbol{\phi}(\ell_p, \nu_p)$.

The local dependence of the AFDM atom on the Doppler parameter can be characterized through its derivative. From (10), define

$$\begin{aligned} \boldsymbol{\psi}(\ell, \nu) &\triangleq \frac{\partial \boldsymbol{\phi}(\ell, \nu)}{\partial \nu}, \\ \psi_{\bar{m}}(\ell, \nu) &= \frac{\partial \phi_{\bar{m}}(\ell, \nu)}{\partial \nu}, \\ \boldsymbol{\Delta}'(\nu) &= \mathrm{diag}(\boldsymbol{d}'(\nu)), \\ \boldsymbol{d}'(\nu) &= \left[-\frac{j2\pi n}{N} e^{-j2\pi\nu n/N} \right]_{n=0}^{N-1}. \end{aligned} \tag{11}$$

A first-order Taylor expansion of the exact atom around its nearest reference grid point then yields

$$\boldsymbol{\phi}(\ell_p, \nu_p) = \boldsymbol{\phi}(\ell_p, \nu_{b_p}) + \varepsilon_p \boldsymbol{\psi}(\ell_p, \nu_{b_p}) + \mathcal{O}(\varepsilon_p^2).$$

This expansion shows that the local atom mismatch depends on the Doppler displacement and the atom sensitivity. The resulting modeling error persists as the noise decreases and can cause an estimation-error floor when a fixed-grid dictionary is used [22].

Reducing the reference grid spacing r_ν can mitigate this approximation error, but it introduces a second difficulty. Because $\boldsymbol{\phi}(\ell, \nu)$ is continuous with respect to ν, two atoms associated with the same delay and sufficiently close Doppler coordinates become highly correlated.

When a physical Doppler shift lies between grid points, the corresponding path response may be approximated more accurately by a combination of neighboring dictionary atoms than by any single grid atom. Because these atoms are highly correlated, different coefficient allocations produce only small differences in the received signal. Such differences can be masked by noise or modeling error, making these allocations difficult to distinguish reliably. Consequently, the energy of a single off-grid path may be distributed among several neighboring recovered coefficients. Interpreting these redundant coefficients as distinct channel paths can overestimate the actual path number. Fig. 2 illustrates the effects of correlation and noise on distinguishing candidate atoms. The centers represent the noiseless responses of the candidate atoms, and the disks indicate the range of possible observations under typical noise perturbations. At low SNR, the responses of highly correlated atoms are sufficiently close that the same noisy observation may be consistent with either atom being active, making it difficult to reliably identify the active atom. Higher SNR improves this distinguishability.

B. Problem Formulation

To account for the Doppler grid mismatch discussed in Section III-A, we retain the integer-delay candidates and treat the Doppler coordinates as unknown continuous parameters.

Combining the integer-delay candidates with the reference Doppler grid gives $K = (\ell_{\max} + 1)M_\nu$ candidate delay–Doppler pairs, which are collected as

$$\mathcal{X} = \{[\ell_g, \nu_g]^T\}_{g=1}^{K}.$$

Here, $g = 1, \ldots, K$ indexes the candidate atoms, $\ell_g \in \mathcal{L}$ is the integer delay of candidate g, and ν_g is its reference Doppler coordinate. The vector $\boldsymbol{\nu} = [\nu_1, \ldots, \nu_K]^T$ collects all such Doppler coordinates.

The reference delay–Doppler grid initializes the K candidate states: it fixes the integer delay coordinate $\ell_g \in \mathcal{L}$ of each candidate, supplies its reference Doppler coordinate, and determines the local refinement scale through the grid spacing r_ν. Thereafter, the Doppler coordinate is continuously refined within the parametric dictionary [29], [31]. Using the unit-gain pilot-response vector in (10) as a dictionary atom, the dictionary is constructed as

$$\boldsymbol{\Phi}(\mathcal{X}) = [\boldsymbol{\phi}(\ell_1, \nu_1), \ldots, \boldsymbol{\phi}(\ell_K, \nu_K)] \in \mathbb{C}^{M \times K}.$$

The observation model in (9) can be expressed in sparse dictionary form as

$$\boldsymbol{y}_T = \boldsymbol{\Phi}(\mathcal{X})\boldsymbol{h} + \boldsymbol{n}, \qquad \boldsymbol{h} = [h_1, \ldots, h_K]^T. \tag{12}$$

The support of $\boldsymbol{h}$ identifies the active candidate paths and their delays, while its nonzero entries represent the corresponding complex path gains. Under ideal matching, each physical path (h_p, ℓ_p, ν_p) corresponds to one active candidate g satisfying $h_g = h_p$ and $(\ell_g, \nu_g) = (\ell_p, \nu_p)$, up to a permutation of the active indices.

Learning $\boldsymbol{\Phi}(\mathcal{X})$ is therefore equivalent to refining only the continuous Doppler coordinates; the fixed-grid model is recovered by evaluating each atom at its assigned reference-grid Doppler coordinate. With the delay coordinates fixed by the reference candidates, joint sparse path recovery and Doppler estimation can be written as

$$\begin{aligned} \min_{\boldsymbol{h},\boldsymbol{\nu}} \quad & \|\boldsymbol{y}_T - \boldsymbol{\Phi}(\mathcal{X})\boldsymbol{h}\|_2^2 \\ \text{s.t.} \quad & \|\boldsymbol{h}\|_0 \le P_{\max}, \\ & \nu_g \in [-\nu_{\max}, \nu_{\max}], \quad g = 1, \dots, K, \end{aligned} \tag{13}$$

where $\boldsymbol{\nu} = [\nu_1, \dots, \nu_K]^T$ and $P_{\max}$ is an upper bound on the number of active candidate paths. Only Dopplers associated with nonzero gains affect the fitted observation; the other grid coordinates initialize possible future paths. Problem (13) expresses joint gain and Doppler estimation by minimizing the observation-fitting error under a sparsity constraint. To address this estimation task, we adopt a Bernoulli–Gaussian model that explicitly represents path existence and gain uncertainty. Marginalizing the active gains yields a Bayesian criterion for support decisions, continuous-Doppler updates, and hyperparameter learning.

IV. Proposed BG-SBL: Path-Support Learning and Continuous Doppler Refinement

In order to alleviate the redundant path representation caused by grid mismatch and highly correlated candidate atoms, this section proposes the Bernoulli–Gaussian sparse Bayesian learning (BG-SBL) method for the joint sparse recovery and parameterized dictionary learning problem established in Section III. This method describes the existence status of candidate paths through Bernoulli variables and adopts a shared Gaussian prior on the active path gains. After marginalizing the path gains, a unified marginal objective is constructed, and the path support, continuous Doppler parameters, and hyperparameters are jointly updated within an expectation/conditional maximization either (ECME) framework.

A. Probabilistic Model and Marginal Objective

To characterize the activity of candidate paths and their complex gains, we introduce a Bernoulli–Gaussian (BG) prior into the parametric observation model in (12) [16], [17]. The BG prior uses a point mass at zero and a conditional Gaussian distribution to describe inactive and active coefficients, respectively, and explicitly controls the prior sparsity level through the Bernoulli activation probability, thereby providing a probabilistic model for joint support inference and gain estimation. Fig. 3 summarizes the hierarchy using the full candidate gain vector $\boldsymbol{h}$ and dictionary $\boldsymbol{\Phi}(\mathcal{X})$.

Let $z_g \in \{0, 1\}$ indicate whether candidate path g is active, and collect all state variables in $\boldsymbol{z} = [z_1, \dots, z_K]^T$. The active-path set and its size are defined as

$$\mathcal{A}(\boldsymbol{z}) \triangleq \{g : z_g = 1\}, \qquad S \triangleq |\mathcal{A}(\boldsymbol{z})|. \tag{14}$$

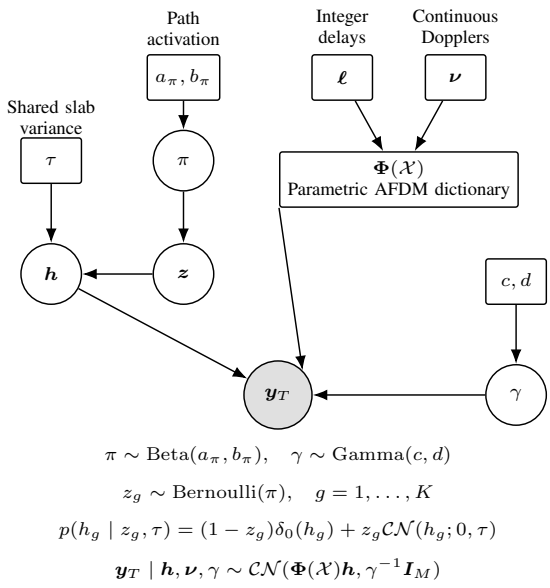


Fig. 3. Bernoulli–Gaussian parametric model.

For an inactive candidate, its gain is exactly zero; for an active candidate, its gain is assigned a zero-mean complex Gaussian prior. Therefore,

$$p(h_g \mid z_g, \tau) = (1 - z_g)\,\delta_0(h_g) + z_g\,\mathcal{CN}(h_g; 0, \tau), \tag{15}$$

where $\delta_0(\cdot)$ denotes a Dirac point mass concentrated at zero, and $\tau > 0$ is the gain prior variance shared by all active paths. The shared variance describes a common prior power scale without requiring the realized gains of different paths to have equal magnitudes. Further assuming that the binary indicators are conditionally independent given a common activation probability π gives

$$p(\boldsymbol{z} \mid \pi) = \pi^S (1 - \pi)^{K - S}, \qquad \pi \sim \mathrm{Beta}(a_\pi, b_\pi), \tag{16}$$

Here, π is the prior probability that a candidate path is active and expresses the channel sparsity in path detection. A principal purpose of the Bernoulli–Gaussian prior is to suppress noise-induced path components at low SNR. When $\pi < 1/2$, activating a new path requires sufficient data evidence, making weak noise-induced matches less likely to be identified as genuine paths. The effective-noise precision γ has the Gamma prior

$$p(\gamma \mid c, d) \propto \gamma^{c-1} e^{-d\gamma}. \tag{17}$$

For a given support and Doppler vector, the zero-gain inactive components disappear from (12). Under the effective white-Gaussian disturbance model in Section II, the likelihood is

$$p(\boldsymbol{y}_T \mid \boldsymbol{h}_{\mathcal{A}}, \gamma, \boldsymbol{z}, \boldsymbol{\nu}) = \mathcal{CN}\big(\boldsymbol{y}_T; \boldsymbol{\Phi}_{\mathcal{A}}(\mathcal{X})\boldsymbol{h}_{\mathcal{A}}, \gamma^{-1}\boldsymbol{I}_M\big), \tag{18}$$

where $\boldsymbol{\Phi}_{\mathcal{A}}(\mathcal{X})$ contains only the atoms indexed by $\mathcal{A}(\boldsymbol{z})$.

To evaluate the support and Doppler coordinates without specifying a particular gain realization, we marginalize the active gains:

$$\begin{aligned} &p(\boldsymbol{y}_T \mid \boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma) \\ &= \int_{\mathbb{C}^S} p(\boldsymbol{y}_T, \boldsymbol{h}_{\mathcal{A}} \mid \boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma)\, \mathrm{d}\boldsymbol{h}_{\mathcal{A}} \\ &= \int_{\mathbb{C}^S} p(\boldsymbol{y}_T \mid \boldsymbol{h}_{\mathcal{A}}, \boldsymbol{z}, \boldsymbol{\nu}, \gamma) p(\boldsymbol{h}_{\mathcal{A}} \mid \tau)\, \mathrm{d}\boldsymbol{h}_{\mathcal{A}} \\ &= \int_{\mathbb{C}^S} \mathcal{CN}\big(\boldsymbol{y}_T; \boldsymbol{\Phi}_{\mathcal{A}}(\mathcal{X})\boldsymbol{h}_{\mathcal{A}}, \gamma^{-1}\boldsymbol{I}_M\big)\, \mathcal{CN}(\boldsymbol{h}_{\mathcal{A}}; \boldsymbol{0}, \tau \boldsymbol{I}_S)\, \mathrm{d}\boldsymbol{h}_{\mathcal{A}} \\ &= \mathcal{CN}(\boldsymbol{y}_T; \boldsymbol{0}, \boldsymbol{C}) = \frac{\exp(-\boldsymbol{y}_T^H \boldsymbol{C}^{-1} \boldsymbol{y}_T)}{\pi^M |\boldsymbol{C}|}. \end{aligned} \tag{19}$$

Equation (19) averages the likelihood over all possible active-gain values using their Gaussian prior as weights and thereby accounts for gain uncertainty. The resulting covariance is

$$\boldsymbol{C}(\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma) = \gamma^{-1}\boldsymbol{I}_M + \tau \boldsymbol{\Phi}_{\mathcal{A}}(\mathcal{X}) \boldsymbol{\Phi}_{\mathcal{A}}^H(\mathcal{X}), \tag{20}$$

where $\gamma^{-1}\boldsymbol{I}_M$ is the effective-disturbance covariance, whereas $\tau \boldsymbol{\Phi}_{\mathcal{A}}(\mathcal{X}) \boldsymbol{\Phi}_{\mathcal{A}}^H(\mathcal{X})$ is the covariance induced by the active-path gains.

After the gains have been marginalized, a candidate model state is specified by $(\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma, \pi)$. Using the conditional-independence structure in Fig. 3, the marginal log-joint criterion can be defined and expanded as

$$\begin{aligned} \mathcal{F}(\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma, \pi) &\triangleq \log p(\boldsymbol{y}_T, \boldsymbol{z}, \pi, \gamma \mid \boldsymbol{\nu}, \tau) \\ &= \log p(\boldsymbol{y}_T \mid \boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma) + \log p(\boldsymbol{z} \mid \pi) \\ &\quad + \log p(\pi) + \log p(\gamma) \\ &= -\log |\boldsymbol{C}| - \boldsymbol{y}_T^H \boldsymbol{C}^{-1} \boldsymbol{y}_T \\ &\quad + S \log \pi + (K - S) \log(1 - \pi) \\ &\quad + (a_\pi - 1) \log \pi + (b_\pi - 1) \log(1 - \pi) \\ &\quad + (c - 1) \log \gamma - d\gamma + \text{const}. \end{aligned} \tag{21}$$

The resulting estimation problem is

$$\begin{aligned} \max_{\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma, \pi} \quad & \mathcal{F}(\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma, \pi) \\ \text{s.t.} \quad & z_g \in \{0, 1\}, \quad g = 1, \ldots, K, \\ & \nu_g \in [-\nu_{\max}, \nu_{\max}], \quad g = 1, \ldots, K, \\ & \tau > 0, \quad \gamma > 0, \quad 0 < \pi < 1. \end{aligned} \tag{22}$$

Equation (22) provides a Bayesian formulation of the joint sparse recovery and dictionary learning task described by (13), where the marginal likelihood characterizes data fidelity and the Bernoulli support prior controls sparsity.

We solve (22) using an ECME framework [25] adapted to the MAP objective. The M-step updates τ and γ by maximizing the expected complete-data log joint density under the retained gain posterior, then conditionally updates the support, Dopplers, and π using $\mathcal{F}$. The subsequent E-step refreshes the gain posterior for the next iteration.

B. ECME Updates

Superscript (t) denotes the state at the end of iteration t, with (0) denoting initialization.

1) E-Step: Active-Gain Posterior: For a given support, Doppler vector, and hyperparameters, the Gaussian likelihood in (18) and the active-gain prior in (15) give the conditional gain posterior

$$\begin{aligned} p(\boldsymbol{h}_{\mathcal{A}} \mid \boldsymbol{y}_T, \boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma) &= \mathcal{CN}(\boldsymbol{h}_{\mathcal{A}}; \boldsymbol{\mu}_{\mathcal{A}}, \boldsymbol{\Sigma}_{\mathcal{A}}), \\ \boldsymbol{\Sigma}_{\mathcal{A}} &= \big(\tau^{-1}\boldsymbol{I}_S + \gamma \boldsymbol{\Phi}_{\mathcal{A}}^H \boldsymbol{\Phi}_{\mathcal{A}}\big)^{-1}, \\ \boldsymbol{\mu}_{\mathcal{A}} &= \gamma \boldsymbol{\Sigma}_{\mathcal{A}} \boldsymbol{\Phi}_{\mathcal{A}}^H \boldsymbol{y}_T, \end{aligned} \tag{23}$$

where $\boldsymbol{\mu}_{\mathcal{A}}$ is the posterior mean and $\boldsymbol{\Sigma}_{\mathcal{A}}$ is the posterior covariance.

The E-step evaluates (23) using the updated support, Dopplers, and hyperparameters to obtain $\boldsymbol{\mu}_{\mathcal{A}}^{(t)}$ and $\boldsymbol{\Sigma}_{\mathcal{A}}^{(t)}$ for the next M-step.

2) M-Step: Parameter Updates: The M-step combines updates of τ and γ based on the retained gain posterior with updates of $\boldsymbol{z}$, $\boldsymbol{\nu}$, and π based on the marginal objective.

Gain-Variance and Noise-Precision Updates. During this update, the incoming support $\mathcal{A}^{(t-1)}$ and its dictionary are fixed, and $S^{(t-1)} = |\mathcal{A}^{(t-1)}|$. The posterior mean $\boldsymbol{\mu}_{\mathcal{A}}^{(t-1)}$ and covariance $\boldsymbol{\Sigma}_{\mathcal{A}}^{(t-1)}$ are taken from the preceding E-step, or from initialization when $t = 1$. Let $\mathcal{E}_h^{(t-1)}$ and $\mathcal{E}_r^{(t-1)}$ denote the posterior expectations of $\|\boldsymbol{h}_{\mathcal{A}}\|_2^2$ and $\|\boldsymbol{y}_T - \boldsymbol{\Phi}_{\mathcal{A}} \boldsymbol{h}_{\mathcal{A}}\|_2^2$, respectively. They are

$$\mathcal{E}_h^{(t-1)} = \|\boldsymbol{\mu}_{\mathcal{A}}^{(t-1)}\|_2^2 + \mathrm{tr}(\boldsymbol{\Sigma}_{\mathcal{A}}^{(t-1)}), \tag{24}$$

$$\mathcal{E}_r^{(t-1)} = \|\boldsymbol{y}_T - \boldsymbol{\Phi}_{\mathcal{A}} \boldsymbol{\mu}_{\mathcal{A}}^{(t-1)}\|_2^2 + \mathrm{tr}(\boldsymbol{\Phi}_{\mathcal{A}} \boldsymbol{\Sigma}_{\mathcal{A}}^{(t-1)} \boldsymbol{\Phi}_{\mathcal{A}}^H). \tag{25}$$

The trace terms account for uncertainty in the active gains.

Taking the expectation under the retained posterior of the complete-data log joint density, including the Gaussian slab prior and the Gamma prior in (17), and omitting terms independent of (τ, γ), gives

$$\begin{aligned} &Q(\tau, \gamma \mid \tau^{(t-1)}, \gamma^{(t-1)}) \\ &\quad = -S^{(t-1)} \log \tau - \frac{\mathcal{E}_h^{(t-1)}}{\tau} \\ &\qquad + (M + c - 1) \log \gamma - \gamma(d + \mathcal{E}_r^{(t-1)}). \end{aligned} \tag{26}$$

There is no factor of $1/2$ because the likelihood and slab prior are proper-complex Gaussian densities. For $S^{(t-1)} > 0$, setting $\partial Q / \partial \tau = 0$ gives

$$\boxed{\tau^{(t)} = \frac{\mathcal{E}_h^{(t-1)}}{S^{(t-1)}}, \qquad S^{(t-1)} > 0.} \tag{27}$$

Likewise, setting $\partial Q / \partial \gamma = 0$ yields

$$\boxed{\gamma^{(t)} = \frac{M + c - 1}{d + \mathcal{E}_r^{(t-1)}}.} \tag{28}$$

If $S^{(t-1)} = 0$, the likelihood does not depend on τ, so its previous positive value is retained until a path is activated.

After τ and γ are updated, the observation covariance inverse is rebuilt for the incoming support and dictionary using the Woodbury identity:

$$\boldsymbol{C}^{-1} = \gamma \boldsymbol{I}_M - \gamma^2 \boldsymbol{\Phi}_{\mathcal{A}} \boldsymbol{\Sigma}_{\mathcal{A}} \boldsymbol{\Phi}_{\mathcal{A}}^H. \tag{29}$$

where $\boldsymbol{\Sigma}_{\mathcal{A}}$ is evaluated by (23) using the updated τ and γ and the incoming active atoms. The required factorization is

$S^{(t-1)} \times S^{(t-1)}$ because inactive gains are zero. The rebuilt inverse is used in the subsequent support and Doppler updates.

Path Decision Criterion. For the support and Doppler updates, $(\tau, \gamma, \pi) = (\tau^{(t)}, \gamma^{(t)}, \pi^{(t-1)})$ are held fixed. Their iteration superscripts are omitted, and $\mathcal{A}$ denotes the current support.

$\Delta_g(\nu)$ denotes the difference in the marginal objective between including and excluding the candidate atom $\boldsymbol{\phi}(\ell_g, \nu)$, with the remaining model state and hyperparameters held fixed. The support update holds these coordinates (ℓ_g, ν), the other candidate states and Dopplers, and the hyperparameters fixed, and compares the marginal objective with this atom included ($z_g = 1$) or excluded ($z_g = 0$):

$$\begin{aligned}\Delta_g(\nu) &\triangleq \mathcal{F}(z_g = 1, \nu_g = \nu) - \mathcal{F}(z_g = 0) \\ &= \log \frac{p(z_g = 1 \mid \boldsymbol{y}_T, \nu_g = \nu, \mathcal{D}_g)}{p(z_g = 0 \mid \boldsymbol{y}_T, \nu_g = \nu, \mathcal{D}_g)} \\ &= \log \frac{\pi}{1-\pi} + \log \frac{p(\boldsymbol{y}_T \mid z_g = 1, \nu_g = \nu, \mathcal{D}_g)}{p(\boldsymbol{y}_T \mid z_g = 0, \nu_g = \nu, \mathcal{D}_g)} \\ &= \log \frac{\pi}{1-\pi} + \mathcal{B}_g(\nu),\end{aligned} \tag{30}$$

where $\mathcal{D}_g = (\boldsymbol{z}_{-g}, \boldsymbol{\nu}_{-g}, \tau, \gamma, \pi)$ collects the fixed quantities, whose arguments in $\mathcal{F}$ are omitted. The term $\log[\pi/(1-\pi)]$ is the prior log-odds, and $\mathcal{B}_g(\nu)$ is the conditional log Bayes factor. An inactive candidate has zero gain, so its Doppler does not affect the marginal likelihood.

Marginalizing the Gaussian gains gives the two conditional observation distributions

$$\begin{aligned}p(\boldsymbol{y}_T \mid z_g = 0, \mathcal{D}_g) &= \mathcal{CN}(\boldsymbol{y}_T; \boldsymbol{0}, \boldsymbol{C}_{-g}), \\ p(\boldsymbol{y}_T \mid z_g = 1, \nu_g = \nu, \mathcal{D}_g) &\\ &= \mathcal{CN}\big(\boldsymbol{y}_T; \boldsymbol{0}, \\ &\qquad \boldsymbol{C}_{-g} + \tau \boldsymbol{\phi}_g(\nu) \boldsymbol{\phi}_g^H(\nu)\big),\end{aligned} \tag{31}$$

where $\boldsymbol{\phi}_g(\nu) \triangleq \boldsymbol{\phi}(\ell_g, \nu)$ is the trial atom and $\boldsymbol{C}_{-g}$ is the covariance contributed by effective noise and the other active paths:

$$\boldsymbol{C}_{-g} = \gamma^{-1} \boldsymbol{I}_M + \tau \sum_{\substack{i \in \mathcal{A} \\ i \neq g}} \boldsymbol{\phi}(\ell_i, \nu_i) \boldsymbol{\phi}^H(\ell_i, \nu_i). \tag{32}$$

Substituting the Gaussian densities in (31) into the log-likelihood ratio cancels their common normalization factor and yields

$$\begin{aligned}\mathcal{B}_g(\nu) &= \log \frac{p(\boldsymbol{y}_T \mid z_g = 1, \nu_g = \nu, \mathcal{D}_g)}{p(\boldsymbol{y}_T \mid z_g = 0, \mathcal{D}_g)} \\ &= -\log \frac{|\boldsymbol{C}_{-g} + \tau \boldsymbol{\phi}_g(\nu) \boldsymbol{\phi}_g^H(\nu)|}{|\boldsymbol{C}_{-g}|} \\ &\quad + \boldsymbol{y}_T^H \Big[\boldsymbol{C}_{-g}^{-1} \\ &\qquad - \big(\boldsymbol{C}_{-g} + \tau \boldsymbol{\phi}_g(\nu) \boldsymbol{\phi}_g^H(\nu)\big)^{-1} \Big] \boldsymbol{y}_T.\end{aligned} \tag{33}$$

The matrix determinant lemma and Sherman–Morrison formula simplify its two terms as

$$\begin{aligned}&\frac{|\boldsymbol{C}_{-g} + \tau \boldsymbol{\phi}_g(\nu) \boldsymbol{\phi}_g^H(\nu)|}{|\boldsymbol{C}_{-g}|} = 1 + \tau s_g(\nu), \\ &\boldsymbol{y}_T^H \Big[\boldsymbol{C}_{-g}^{-1} \\ &\quad - \big(\boldsymbol{C}_{-g} + \tau \boldsymbol{\phi}_g(\nu) \boldsymbol{\phi}_g^H(\nu)\big)^{-1} \Big] \boldsymbol{y}_T \\ &\qquad = \frac{\tau |\boldsymbol{\phi}_g^H(\nu) \boldsymbol{C}_{-g}^{-1} \boldsymbol{y}_T|^2}{1 + \tau s_g(\nu)} = \frac{\tau |q_g(\nu)|^2}{1 + \tau s_g(\nu)},\end{aligned}$$

where

$$\begin{aligned}s_g(\nu) &\triangleq \boldsymbol{\phi}_g^H(\nu) \boldsymbol{C}_{-g}^{-1} \boldsymbol{\phi}_g(\nu), \\ q_g(\nu) &\triangleq \boldsymbol{\phi}_g^H(\nu) \boldsymbol{C}_{-g}^{-1} \boldsymbol{y}_T\end{aligned} \tag{34}$$

In (34), $s_g(\nu)$ denotes the covariance-weighted atom energy, and $q_g(\nu)$ denotes the covariance-weighted correlation between the atom and the observation.

For $g \in \mathcal{A}$, the inverse in (34) is obtained by removing the current covariance contribution of candidate g:

$$\begin{aligned}\boldsymbol{C}_{-g}^{-1} &= \boldsymbol{C}^{-1} \\ &\quad + \frac{\tau \boldsymbol{C}^{-1} \boldsymbol{\phi}_g(\nu_g) \boldsymbol{\phi}_g^H(\nu_g) \boldsymbol{C}^{-1}}{1 - \tau \boldsymbol{\phi}_g^H(\nu_g) \boldsymbol{C}^{-1} \boldsymbol{\phi}_g(\nu_g)}.\end{aligned} \tag{35}$$

Substituting the determinant ratio and quadratic-form difference into (33) gives

$$\mathcal{B}_g(\nu) = \frac{\tau |q_g(\nu)|^2}{1 + \tau s_g(\nu)} - \log[1 + \tau s_g(\nu)], \tag{36}$$

where the first term rewards additional data fit and the second penalizes the covariance expansion caused by adding the candidate.

Combining (30) and (36) gives the complete decision score

$$\begin{aligned}\Delta_g(\nu) = \log \frac{\pi}{1-\pi} &+ \frac{\tau |q_g(\nu)|^2}{1 + \tau s_g(\nu)} \\ &- \log[1 + \tau s_g(\nu)].\end{aligned}$$

Since $\Delta_g(\nu)$ is the objective difference and the conditional posterior log-odds, the conditional MAP decision [26] is

$$z_g \leftarrow \begin{cases} 1, & \Delta_g(\nu) \geq 0, \\ 0, & \Delta_g(\nu) < 0. \end{cases} \tag{37}$$

For each evaluated candidate, we compute $\Delta_g(\nu)$ and update z_g according to (37). A transition from zero to one adds the candidate to $\mathcal{A}$, a transition from one to zero removes it, and an unchanged state preserves its support membership. Each support update does not decrease $\mathcal{F}$.

When a candidate transitions from inactive to active, adding its atom at the evaluated Doppler coordinate ν updates the covariance inverse as

$$\boldsymbol{C}_{\text{new}}^{-1} = \boldsymbol{C}_{-g}^{-1} - \frac{\tau \boldsymbol{C}_{-g}^{-1} \boldsymbol{\phi}(\ell_g, \nu) \boldsymbol{\phi}^H(\ell_g, \nu) \boldsymbol{C}_{-g}^{-1}}{1 + \tau \boldsymbol{\phi}^H(\ell_g, \nu) \boldsymbol{C}_{-g}^{-1} \boldsymbol{\phi}(\ell_g, \nu)}. \tag{38}$$

Deleting a path retains $\boldsymbol{C}_{-g}^{-1}$ from (35) as the new inverse. Each addition or removal changes the covariance by a rank-one term and avoids a new observation-domain factorization.

The vector $\boldsymbol{C}^{-1}\boldsymbol{y}_T$ is updated together with the inverse. The updated inverse is used to compute the decision score for the next candidate.

The active-path support pass evaluates each currently active candidate at its retained Doppler coordinate. If a candidate is deleted, its Doppler is reset to its initial Doppler-grid coordinate $\nu_g^{(0)}$ for later reevaluation.

Doppler Refinement. After the active-path support pass, the Doppler of each retained path is refined by maximizing $\mathcal{B}_g(\nu)$ over a moving local interval, with the support fixed.

The local search uses the initial Doppler-grid spacing r_ν as its scale and is centered at the current Doppler estimate. At iteration t, the search interval is defined as

$$\mathcal{I}_g^{(t)} = \left[\nu_g^{(t-1)} - \frac{r_\nu}{2}, \nu_g^{(t-1)} + \frac{r_\nu}{2}\right] \cap [-\nu_{\max}, \nu_{\max}]. \quad (39)$$

where $\nu_g^{(t-1)}$ is the retained estimate from the preceding iteration and r_ν is the grid spacing defined in Section III. The interval permits a local correction of at most half a grid spacing per iteration and moves with the updated estimate. During a single Doppler-coordinate update, the support is fixed. Therefore, the Bernoulli terms in (21), the priors on π and γ, and every contribution from the other coordinates are constant with respect to ν_g. The only term that varies with ν_g is $\mathcal{B}_g(\nu)$ in (36). Hence

$$\boxed{\nu_g^\star \in \underset{\nu\in\mathcal{I}_g^{(t)}}{\arg\max}\, \mathcal{B}_g(\nu) = \underset{\nu\in\mathcal{I}_g^{(t)}}{\arg\max}\, \mathcal{F}.} \quad (40)$$

We solve the local Doppler update using a safeguarded Newton iteration:

$$\widetilde{\nu}_g^{(j+1)} = \Pi_{\mathcal{I}_g^{(t)}}\left[\nu_g^{(j)} - \alpha_j \frac{\mathcal{B}_g'\left(\nu_g^{(j)}\right)}{\mathcal{B}_g''\left(\nu_g^{(j)}\right)}\right], \quad (41)$$

where j indexes the inner Newton iterations, $\Pi_{\mathcal{I}_g^{(t)}}(\cdot)$ clips a proposal to the search interval, and α_j is a backtracking step size initialized to one and successively halved. A Newton update is attempted only when $\mathcal{B}_g''(\nu_g^{(j)}) < 0$. The first candidate that strictly increases $\mathcal{B}_g$ is accepted as $\nu_g^{(j+1)}$; if the curvature condition is not satisfied or no improving candidate is found, the current estimate is retained. Thus, each accepted update increases the marginal objective $\mathcal{F}$.

With $\boldsymbol{C}_{-g}$ and the hyperparameters fixed, the analytic derivatives are

$$\begin{aligned}
\mathcal{B}_g' &= \frac{\tau(r_g' - s_g')}{D_g} - \frac{\tau^2 r_g s_g'}{D_g^2}, \\
\mathcal{B}_g'' &= \frac{\tau(r_g'' - s_g'')}{D_g} + \frac{\tau^2\left[(s_g')^2 - 2r_g' s_g' - r_g s_g''\right]}{D_g^2} \\
&\quad + \frac{2\tau^3 r_g (s_g')^2}{D_g^3}.
\end{aligned} \quad (42)$$

where $D_g = 1 + \tau s_g(\nu)$, $r_g(\nu) = |q_g(\nu)|^2$, and primes denote differentiation with respect to ν. The required derivatives of s_g and r_g are

$$\begin{aligned}
s_g' &= 2\,\mathrm{Re}\left\{\boldsymbol{\psi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{\phi}_g\right\}, \\
s_g'' &= 2\,\mathrm{Re}\left\{\boldsymbol{\chi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{\phi}_g + \boldsymbol{\psi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{\psi}_g\right\}, \\
r_g' &= 2\,\mathrm{Re}\left\{q_g^* \boldsymbol{\psi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{y}_T\right\}, \\
r_g'' &= 2\,\mathrm{Re}\left\{\left|\boldsymbol{\psi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{y}_T\right|^2 + q_g^* \boldsymbol{\chi}_g^H \boldsymbol{C}_{-g}^{-1}\boldsymbol{y}_T\right\}.
\end{aligned} \quad (43)$$

where $\boldsymbol{\psi}_g = \partial\boldsymbol{\phi}_g(\nu)/\partial\nu$ and $\boldsymbol{\chi}_g = \partial^2\boldsymbol{\phi}_g(\nu)/\partial\nu^2$. After the Doppler refinement of each active path, its old atom is removed using (35), and the refined atom is added using (38) to update the covariance inverse.

Path Search and Addition. After the activity states of the other candidates and the hyperparameters are updated, previously inactive candidates may satisfy the activation criterion and therefore need to be reevaluated. Let $\mathcal{A}^c = \{g : z_g = 0\}$ denote the index set of currently inactive candidates. To reduce the computation time of continuous Doppler search, we first rank the inactive candidates and perform Taylor–Newton search only on the top K_b candidates, where K_b is defined as

$$K_b = \max\{K_{\max} - |\mathcal{A}|, 0\}, \qquad K_{\max} = \left\lfloor \frac{M}{\ln K} \right\rfloor, \quad (44)$$

where M is the number of observations and K is the number of initial delay–Doppler candidates. $K_{\max}$ denotes the upper limit on the total number of currently active paths and inactive candidates to be searched in the current iteration. Its value is chosen based on the number of recoverable sparse components determined by the number of observations and the dictionary size in [23]. At each iteration, after the active candidates have been processed, candidates in $\mathcal{A}^c$ are ranked by $\mathcal{B}_g(\nu_g^{(0)})$. These scores are recomputed at every iteration using the current support and hyperparameters. The top K_b candidates are selected for Taylor–Newton search and support reevaluation. If $K_b = 0$, this search is skipped.

For each selected inactive candidate, the Doppler search starts from its initial grid coordinate and uses the interval in (39) centered at $\nu_g^{(0)}$. Since this coordinate may differ from the Doppler of a physical path and Newton's method is sensitive to initialization, we first use a first-order Taylor approximation to find a suitable starting point within the local interval and then apply Newton's method to optimize the exact objective. The support decision score $\Delta_g(\nu)$ is evaluated at the refined Doppler coordinate using (30), and candidate activation follows (37).

Therefore, every active atom must undergo a Taylor-Newton search upon activation. If the active atom is retained in subsequent iterations, Newton refinement proceeds directly from this estimate, eliminating the need to re-execute the Taylor prediction. This also saves computation time.

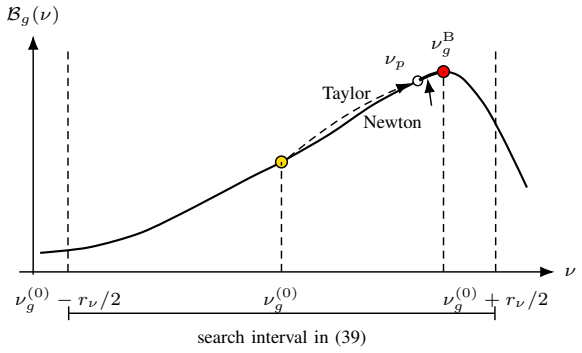


Fig. 4. Taylor–Newton Doppler search for a selected inactive candidate.

For each selected candidate $g \in \mathcal{A}^c$, a first-order Taylor approximation around its initial Doppler coordinate $\nu_g^{(0)}$ is used to obtain a starting point ν_p for Newton refinement within the local search interval, as illustrated in Fig. 4:

$$\boldsymbol{\phi}(\ell_g, \nu) \simeq \boldsymbol{\phi}(\ell_g, \nu_g^{(0)}) + (\nu - \nu_g^{(0)}) \left.\frac{\partial \boldsymbol{\phi}(\ell_g, \nu)}{\partial \nu}\right|_{\nu=\nu_g^{(0)}}.$$

Substituting the expanded atom into (34) gives $s_{T,g}(\nu)$ and $|q_{T,g}(\nu)|^2$, both of which are quadratic in $\nu - \nu_g^{(0)}$. Substitution into (36) then gives

$$\ell_{T,g}(\nu) = \frac{\tau |q_{T,g}(\nu)|^2}{1 + \tau s_{T,g}(\nu)} - \log[1 + \tau s_{T,g}(\nu)]\,.$$

Setting its derivative to zero and clearing the denominator gives a cubic polynomial. Let $\mathcal{R}_g$ contain its real roots in $\mathcal{I}_g^{(t)}$. The Taylor predictor is selected from

$$\boxed{\nu_p \in \arg\max_{\nu \in \mathcal{V}_g} \ell_{T,g}(\nu), \qquad \mathcal{V}_g = \begin{cases} \mathcal{R}_g, & \mathcal{R}_g \neq \varnothing, \\ \partial \mathcal{I}_g^{(t)}, & \mathcal{R}_g = \varnothing, \end{cases}} \tag{45}$$

where $\partial \mathcal{I}_g^{(t)}$ contains the two interval endpoints. Thus, the endpoints are evaluated only when the cubic equation has no feasible real stationary point. Starting from ν_p, the Newton update in (41) refines the Doppler to ν_g^{B}. The candidate is then reevaluated using the exact atom $\boldsymbol{\phi}(\ell_g, \nu_g^{\mathrm{B}})$ and the support rule in (37). Once a candidate is activated, the covariance inverse is updated by (38), and its Doppler is refined in subsequent iterations using the active-path Doppler refinement described above.

Activation-Probability Update. With the updated support $\mathcal{A}^{(t)}$ fixed, $S^{(t)} = |\mathcal{A}^{(t)}|$ of the K candidate paths are active. Their binary indicators give the Bernoulli likelihood $p(\boldsymbol{z} \mid \pi) = \pi^{S^{(t)}}(1-\pi)^{K-S^{(t)}}$. Combining this likelihood with the Beta prior in (16) gives

$$p(\pi \mid \boldsymbol{z}) \propto \pi^{S^{(t)}+a_\pi-1}(1-\pi)^{K-S^{(t)}+b_\pi-1}.$$

Thus, the conditional posterior density $p(\pi \mid \boldsymbol{z})$ is a Beta density with parameters $S^{(t)}+a_\pi$ and $K-S^{(t)}+b_\pi$. For $a_\pi > 1$ and $b_\pi > 1$, maximizing this density gives the conditional maximum-a-posteriori (MAP) update [32]:

$$\boxed{\begin{aligned} \pi^{(t)} &= \arg\max_{0<\pi<1} p(\pi \mid \boldsymbol{z}) \\ &= \frac{S^{(t)} + a_\pi - 1}{K + a_\pi + b_\pi - 2}. \end{aligned}} \tag{46}$$

Because $a_\pi, b_\pi > 1$, the estimate remains in $(0,1)$ even if $S^{(t)} = 0$ or $S^{(t)} = K$. The updated $\pi^{(t)}$ enters the next iteration through the prior-odds term $\log[\pi/(1-\pi)]$ in (30).

TABLE I
COMPUTATIONAL COMPLEXITY OF THE COMPARED ESTIMATORS

Estimator	Dominant arithmetic cost
EPA-DR	$\mathcal{O}[N(G+1)]$
OG-SBL	$\mathcal{O}\big(M^3 + M^2K + MK^2 + K^3 + J_{\mathrm{OG}}K^2\big)$
GR-SBL	$\mathcal{O}\big(M^3 + M^2K + MK^2 + \bar{P}[M^3 + M^2K + N_\Delta M^2]\big)$
GE-SBL	$\mathcal{O}\big(M^3 + M^2K + MK^2 + MK\bar{P} + K\bar{P}^2 + \bar{P}^3\big)$
BG-SBL	$\mathcal{O}\big(M^2(K+K_b)$ $+ (MS+S^2)(N_{\mathrm{N}}S + N_{\mathrm{B}}K_b) + MS^2 + S^3\big)$

SBL costs are per outer iteration; EPA-DR is per complete channel estimate.

C. Algorithm Summary and Complexity

1) Algorithm Summary: Algorithm 1 implements an M–E cycle, using the gain posterior retained from the preceding iteration for the M-step. The initial candidate set is $\mathcal{X}^{(0)} = \{[\ell_g, \nu_g^{(0)}]^T\}_{g=1}^K$ from Section III, with each atom defined by (10). The algorithm returns the final support, path gains, delays, Dopplers, and reconstructed channel.

The algorithm stops when the support is unchanged and the following parameter-change measure falls below ϵ, or when $t = T_{\max}$:

$$\begin{aligned} \epsilon^{(t)} = \max\Bigg\{ &\max_{g \in \mathcal{A}^{(t)}} \frac{|\nu_g^{(t)} - \nu_g^{(t-1)}|}{r_\nu}, \\ &\frac{|\tau^{(t)} - \tau^{(t-1)}|}{\max\{\tau^{(t-1)}, \epsilon_0\}}, \\ &\frac{|\gamma^{(t)} - \gamma^{(t-1)}|}{\max\{\gamma^{(t-1)}, \epsilon_0\}} \Bigg\}. \end{aligned} \tag{47}$$

Here, $\boldsymbol{z}^{(t)} = \boldsymbol{z}^{(t-1)}$ checks support stability. The three terms in (47) measure Doppler changes relative to r_ν and relative changes in τ and γ; ϵ_0 prevents division by a near-zero value. The inner maximum is zero if $\mathcal{A}^{(t)} = \varnothing$. Since (46) depends only on the support size, no separate check for π is needed.

2) Complexity Analysis: Let $S = |\mathcal{A}|$ and K_b denote the current active-set size and the number of inactive candidates selected for continuous Doppler search, respectively. The gain posterior in OG-SBL, GR-SBL, and centralized GE-SBL contains all K candidate gains [21], [23], whereas BG-SBL factors only the S active gains at a cost of $\mathcal{O}(MS^2 + S^3)$. Table I summarizes the dominant arithmetic costs of the compared estimators.

For Algorithm 1, active-path maintenance, inactive screening, and inverse reconstruction cost $\mathcal{O}(M^2S)$, $\mathcal{O}[M^2(K-S)]$, and $\mathcal{O}(M^2S)$, respectively. Let N_{N} and N_{B} denote the

Algorithm 1 BG-SBL for AFDM Channel Estimation

Input: $\boldsymbol{y}_T$, initial candidate set $\mathcal{X}^{(0)}$, r_ν, $\nu_{\max}$, (a_π, b_π), (c, d), tolerances (ϵ, ϵ_0), maximum iteration number $T_{\max}$
Output: $\widehat{P}$, $\widehat{\boldsymbol{h}}$, $\widehat{\boldsymbol{\ell}}$, $\widehat{\nu}$, $\widehat{\boldsymbol{H}}$

Form $\boldsymbol{\Phi}(\mathcal{X}^{(0)})$; initialize $\boldsymbol{z} = \boldsymbol{0}$ and $\boldsymbol{\nu} = \boldsymbol{\nu}^{(0)}$
Initialize $\tau > 0$, $\gamma > 0$, and $0 < \pi < 1$
Set $K_{\max} = \lfloor M / \ln K \rfloor$ by (44)
Initialize empty $\boldsymbol{\mu}_{\mathcal{A}}$ and $\boldsymbol{\Sigma}_{\mathcal{A}}$
for $t = 1, \ldots, T_{\max}$ **do**
  Store $(\boldsymbol{z}, \boldsymbol{\nu}, \tau, \gamma)$ from iteration $t-1$
  M-step: parameter updates
  Gain-variance and noise-precision updates
  With the current support and dictionary fixed, compute $\mathcal{E}_h$ and $\mathcal{E}_r$ from the incoming gain posterior by (24)–(25)
  Update τ and γ by (27) and (28); retain τ if $|\mathcal{A}| = 0$
  Rebuild $\boldsymbol{C}^{-1}$ by (29) using the updated (τ, γ); recompute $\boldsymbol{C}^{-1}\boldsymbol{y}_T$
  Path retention and deletion
  for all $g \in \mathcal{A}$ **do**
    Form $\boldsymbol{C}_{-g}^{-1}$ by (35); retain or delete g by (37)
    If deleted, reset $\nu_g \leftarrow \nu_g^{(0)}$, set $\boldsymbol{C}^{-1} \leftarrow \boldsymbol{C}_{-g}^{-1}$, and recompute $\boldsymbol{C}^{-1}\boldsymbol{y}_T$
  end for
  Doppler refinement
  for all retained $g \in \mathcal{A}$ **do**
    Form $\boldsymbol{C}_{-g}^{-1}$ by (35); update ν_g by (41)
    Update $\boldsymbol{C}^{-1}$ by (38) and recompute $\boldsymbol{C}^{-1}\boldsymbol{y}_T$
  end for
  Path search and addition
  Set $K_b = \max\{K_{\max} - |\mathcal{A}|, 0\}$; rank all $g \in \mathcal{A}^{\mathrm{c}}$ by $\mathcal{B}_g(\nu_g^{(0)})$ and select the top K_b
  for all selected $g \in \mathcal{A}^{\mathrm{c}}$ **do**
    Use $\boldsymbol{C}_{-g}^{-1} = \boldsymbol{C}^{-1}$; find ν_p by (45), then ν_g^{B} by (41)
    Test activation by (37); if accepted, set $\nu_g \leftarrow \nu_g^{\mathrm{B}}$, update $\boldsymbol{C}^{-1}$ by (38), and recompute $\boldsymbol{C}^{-1}\boldsymbol{y}_T$
  end for
  Activation-probability update
  Set $S = |\mathcal{A}|$ after the support updates; update π by (46)
  E-step: active-gain posterior refresh
  Compute $(\boldsymbol{\mu}_{\mathcal{A}}, \boldsymbol{\Sigma}_{\mathcal{A}})$ by (23) using the updated support, Dopplers, τ, and γ; save this posterior for the next M-step
  Compute $\epsilon^{(t)}$ by (47)
  if $\epsilon^{(t)} < \epsilon$ and $\boldsymbol{z}$ is unchanged **then**
    break
  end if
end for
Set $\widehat{\mathcal{A}} = \mathcal{A}(\boldsymbol{z})$, $\widehat{P} = |\widehat{\mathcal{A}}|$, and $\widehat{\boldsymbol{h}} = \boldsymbol{\mu}_{\widehat{\mathcal{A}}}$ from the final E-step
Set $\widehat{\boldsymbol{\ell}} = \boldsymbol{\ell}_{\widehat{\mathcal{A}}}$ and $\widehat{\boldsymbol{\nu}} = \boldsymbol{\nu}_{\widehat{\mathcal{A}}}$
Reconstruct $\widehat{\boldsymbol{H}}$ by (3c)

Newton iteration counts for an active and a selected inactive candidate. Their Doppler searches cost $\mathcal{O}[N_{\mathrm{N}}(MS + S^2)S]$ and $\mathcal{O}[N_{\mathrm{B}}(MS + S^2)K_b]$; accepted activations cost at most $\mathcal{O}(M^2K_b)$. Combining these terms gives $\mathcal{O}\{M^2(K + K_b) + (MS + S^2)(N_{\mathrm{N}}S + N_{\mathrm{B}}K_b) + MS^2 + S^3\}$ per iteration. Screening still covers all inactive atoms; continuous search covers only the K_b shortlisted atoms.

The OG-SBL, GR-SBL, and GE-SBL implementations share an $\mathcal{O}(M^3+M^2K+MK^2)$ observation-domain posterior cost, with additional method-specific work. In their complexity expressions, $\bar{P}$ is the number of paths selected for Doppler refinement, N_Δ is the number of Doppler points tested per GR-SBL path, and J_{OG} is the number of OG-SBL coordinate sweeps.

For the SISO EPA-DR baseline, threshold detection processes $G + 1$ received pilot samples. Each retained sample initializes one diagonal of the equivalent channel matrix, and reconstructing its remaining entries requires $N - 1$ complex multiplications [11]. The reconstruction therefore requires at most $(G + 1)(N - 1)$ complex multiplications, giving an arithmetic cost of $\mathcal{O}[N(G+1)]$ per complete channel estimate.

Compared with OG-SBL, GR-SBL, and GE-SBL, BG-SBL reduces the gain posterior dimension from K to S and limits continuous Doppler searches of inactive candidates to K_b candidates. Rank-one updates avoid repeated covariance matrix factorizations, while Newton refinement avoids GR-SBL's dense Doppler grid search. BG-SBL therefore reduces posterior inference, covariance update, and Doppler optimization costs, enabling joint estimation of path support, gains, and continuous Dopplers at lower computational cost.

V. Cramér–Rao Bound Analysis

This section derives the Cramér–Rao bound (CRB) for complex path gains and continuous Dopplers under a colored-Gaussian approximate observation model, assuming that the true path number P and integer delays $(\ell_1, \ldots, \ell_P)$ are known. The derivation incorporates residual data interference into the observation covariance and treats the thermal-noise variance as an unknown nuisance parameter. The resulting CRB provides a performance reference for the estimation accuracy of gains and continuous Dopplers.

1) Parameter Definitions and Observation Model: Let $\boldsymbol{h} = [h_1, \ldots, h_P]^T$ and $\boldsymbol{\nu} = [\nu_1, \ldots, \nu_P]^T$ denote the path gains and continuous normalized Dopplers. With $\boldsymbol{h}_R = \mathrm{Re}\{\boldsymbol{h}\}$ and $\boldsymbol{h}_I = \mathrm{Im}\{\boldsymbol{h}\}$, define

$$\boldsymbol{\theta} = [\boldsymbol{h}_R^T, \boldsymbol{h}_I^T, \boldsymbol{\nu}^T]^T, \qquad \boldsymbol{\theta}_f = [\boldsymbol{\theta}^T, \sigma_n^2]^T \in \mathbb{R}^{3P+1}. \quad (48)$$

Here, $\boldsymbol{\theta} \in \mathbb{R}^{3P}$ contains the path gains and Dopplers of interest, while $\boldsymbol{\theta}_f$ is the full unknown-parameter vector, including the thermal-noise variance σ_n^2. Since the observation covariance depends on both the path parameters and the unknown thermal-noise variance, we first construct the joint Fisher information matrix for $\boldsymbol{\theta}_f$. The CRB for $\boldsymbol{\theta}$ is then obtained by eliminating the noise variance as a nuisance parameter.

Let $\boldsymbol{H}_T \in \mathbb{C}^{M \times N}$ denote the channel response over the truncated receive window, with $H_T[\bar{m}, m] = H[\widetilde{m}, m]$ for $\bar{m} = 0, \ldots, M-1$ and $m = 0, \ldots, N-1$, where $\widetilde{m}$ is defined below (10). The retained observations contain residual data interference $\boldsymbol{H}_T(\boldsymbol{\theta})\boldsymbol{s}_d$ despite the pilot guard zones. Its covariance can be correlated and depend on $\boldsymbol{\theta}$. With $\boldsymbol{C}_d = \mathbb{E}[\boldsymbol{s}_d\boldsymbol{s}_d^H]$ and $\boldsymbol{\Phi} = [\boldsymbol{\phi}(\ell_1, \nu_1), \ldots, \boldsymbol{\phi}(\ell_P, \nu_P)]$, $\boldsymbol{C}_d$ is diagonal with entries P_d on $\mathcal{D}$ and zero elsewhere. Matching the first two moments gives the colored-Gaussian approximate observation model [30]

$$\begin{aligned} p_{\mathrm{G}}(\boldsymbol{y}_T; \boldsymbol{\theta}_f) &= \mathcal{CN}(\boldsymbol{y}_T; \boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{C}(\boldsymbol{\theta}_f)), \\ \boldsymbol{\mu}(\boldsymbol{\theta}) &= \boldsymbol{\Phi}\boldsymbol{h} = \boldsymbol{H}_T(\boldsymbol{\theta})\boldsymbol{s}_p, \\ \boldsymbol{C}(\boldsymbol{\theta}_f) &= \sigma_n^2\boldsymbol{I}_M + \boldsymbol{H}_T(\boldsymbol{\theta})\boldsymbol{C}_d\boldsymbol{H}_T^H(\boldsymbol{\theta}). \end{aligned} \quad (49)$$

For finite-alphabet data such as QPSK, the exact marginalized likelihood is a Gaussian mixture; hence the bound below pertains to (49).

2) Joint Fisher Information Matrix: For unbiased estimators of the full parameter vector $\boldsymbol{\theta}_f$, the proper-complex Gaussian

model gives the joint CRB and Fisher information matrix (FIM):

$$\mathrm{Cov}(\widehat{\boldsymbol{\theta}}_f) \succeq \boldsymbol{J}_f^{-1},$$
$$\boldsymbol{J}_f = \mathbb{E}_{\boldsymbol{y}_T \sim p_{\mathrm{G}}(\cdot;\boldsymbol{\theta}_f)}\Big[-\nabla_{\boldsymbol{\theta}_f}\nabla_{\boldsymbol{\theta}_f}^T \log p_{\mathrm{G}}(\boldsymbol{y}_T;\boldsymbol{\theta}_f)\Big]. \quad (50)$$

For $i = 1,\dots,3P$, write $\boldsymbol{\mu}_i = \partial\boldsymbol{\mu}/\partial\theta_i$, $\boldsymbol{C}_i = \partial\boldsymbol{C}/\partial\theta_i$, and $\boldsymbol{H}_i = \partial\boldsymbol{H}/\partial\theta_i$, with $\boldsymbol{H}_{T,i} = \partial\boldsymbol{H}_T/\partial\theta_i$. The covariance derivative is

$$\boldsymbol{C}_i = \boldsymbol{H}_{T,i}\boldsymbol{C}_d\boldsymbol{H}_T^H + \boldsymbol{H}_T\boldsymbol{C}_d\boldsymbol{H}_{T,i}^H. \quad (51)$$

Differentiating the AFDM channel matrix in (3c) with respect to the real and imaginary parts of the path gains and the continuous Dopplers gives

$$\frac{\partial\boldsymbol{H}}{\partial h_{R,p}} = \boldsymbol{A}\boldsymbol{\Delta}(\nu_p)\boldsymbol{\Pi}^{\ell_p}\boldsymbol{A}^H,$$
$$\frac{\partial\boldsymbol{H}}{\partial h_{I,p}} = j\boldsymbol{A}\boldsymbol{\Delta}(\nu_p)\boldsymbol{\Pi}^{\ell_p}\boldsymbol{A}^H,$$
$$\frac{\partial\boldsymbol{H}}{\partial \nu_p} = h_p\boldsymbol{A}\boldsymbol{\Delta}'(\nu_p)\boldsymbol{\Pi}^{\ell_p}\boldsymbol{A}^H. \quad (52)$$

With $\boldsymbol{\phi}_p = \boldsymbol{\phi}(\ell_p,\nu_p)$ and $\boldsymbol{\psi}_p = \partial\boldsymbol{\phi}(\ell_p,\nu_p)/\partial\nu_p$ from (11), the mean derivatives are

$$\frac{\partial\boldsymbol{\mu}}{\partial h_{R,p}} = \boldsymbol{\phi}_p, \quad \frac{\partial\boldsymbol{\mu}}{\partial h_{I,p}} = j\boldsymbol{\phi}_p, \quad \frac{\partial\boldsymbol{\mu}}{\partial \nu_p} = h_p\boldsymbol{\psi}_p. \quad (53)$$

Also, $\partial\boldsymbol{\mu}/\partial\sigma_n^2 = \boldsymbol{0}$ and $\partial\boldsymbol{C}/\partial\sigma_n^2 = \boldsymbol{I}_M$.

Substituting the covariance and mean derivatives in (51) and (53), together with the derivatives with respect to the noise variance, into the general FIM formula for a proper-complex Gaussian model [32] gives the blocks of the joint FIM:

$$[\boldsymbol{J}_{\theta\theta}]_{i,j} = 2\,\mathrm{Re}\big\{\boldsymbol{\mu}_i^H\boldsymbol{C}^{-1}\boldsymbol{\mu}_j\big\} + \mathrm{tr}\big(\boldsymbol{C}^{-1}\boldsymbol{C}_i\boldsymbol{C}^{-1}\boldsymbol{C}_j\big),$$
$$[\boldsymbol{j}_{\theta,\sigma_n^2}]_i = \mathrm{tr}\big(\boldsymbol{C}^{-1}\boldsymbol{C}_i\boldsymbol{C}^{-1}\big),$$
$$J_{\sigma_n^2,\sigma_n^2} = \mathrm{tr}\big(\boldsymbol{C}^{-2}\big). \quad (54)$$

Here $i,j = 1,\dots,3P$. Thus, the complete FIM has the block form

$$\boldsymbol{J}_f = \begin{bmatrix} \boldsymbol{J}_{\theta\theta} & \boldsymbol{j}_{\theta,\sigma_n^2} \\ \boldsymbol{j}_{\theta,\sigma_n^2}^T & J_{\sigma_n^2,\sigma_n^2} \end{bmatrix}. \quad (55)$$

3) CRBs for Path Gains and Dopplers: Because residual data interference makes $\boldsymbol{C}$ depend on $\boldsymbol{\theta}$, the cross-information vector $\boldsymbol{j}_{\theta,\sigma_n^2}$ is generally nonzero, coupling the path parameters with the thermal-noise variance. Taking the Schur complement of the noise-variance block accounts for this coupling and gives the effective information for $\boldsymbol{\theta}$. Its inverse equals the $\boldsymbol{\theta}$ block of $\boldsymbol{J}_f^{-1}$ and gives the joint CRB for the path gains and Dopplers:

$$\overline{\boldsymbol{J}}_\theta = \boldsymbol{J}_{\theta\theta} - \boldsymbol{j}_{\theta,\sigma_n^2}J_{\sigma_n^2,\sigma_n^2}^{-1}\boldsymbol{j}_{\theta,\sigma_n^2}^T,$$
$$\mathrm{Cov}(\widehat{\boldsymbol{\theta}}) \succeq [\boldsymbol{J}_f^{-1}]_{\theta\theta} = \overline{\boldsymbol{J}}_\theta^{-1}. \quad (56)$$

For Doppler estimation, the gains are also nuisance parameters. With $\boldsymbol{a} = [\boldsymbol{h}_R^T, \boldsymbol{h}_I^T]^T$, a second Schur complement gives

$$\boldsymbol{J}_\nu^{\mathrm{e}} = \overline{\boldsymbol{J}}_{\nu\nu} - \overline{\boldsymbol{J}}_{\nu a}\overline{\boldsymbol{J}}_{aa}^{-1}\overline{\boldsymbol{J}}_{a\nu}. \quad (57)$$

The inverse $(\boldsymbol{J}_\nu^{\mathrm{e}})^{-1}$ is the Doppler CRB with both the path gains and thermal-noise variance unknown.

4) Doppler RMSE and Channel NMSE Bounds: For unbiased Doppler estimates, the mean squared error averaged over the P paths satisfies

$$\mathrm{RMSE}_\nu^2 = \frac{1}{P}\sum_{p=1}^{P}\mathbb{E}\big[(\widehat{\nu}_p - \nu_p)^2\big]$$
$$= \frac{1}{P}\,\mathrm{tr}(\mathrm{Cov}(\widehat{\boldsymbol{\nu}}))$$
$$\geq \frac{1}{P}\,\mathrm{tr}\big((\boldsymbol{J}_\nu^{\mathrm{e}})^{-1}\big).$$

Taking the square root gives the Doppler RMSE bound used in the simulations:

$$\mathrm{RMSE}_{\nu,\mathrm{CRB}} = \sqrt{\frac{1}{P}\,\mathrm{tr}((\boldsymbol{J}_\nu^{\mathrm{e}})^{-1})}. \quad (58)$$

To transform the joint CRB for $\boldsymbol{\theta}$ into a bound on the reconstructed AFDM channel, define the real-valued channel vector and its Jacobian as

$$\boldsymbol{g}_H(\boldsymbol{\theta}) = \begin{bmatrix}\mathrm{Re}\{\mathrm{vec}(\boldsymbol{H})\} \\ \mathrm{Im}\{\mathrm{vec}(\boldsymbol{H})\}\end{bmatrix},$$
$$\boldsymbol{G}_H = \frac{\partial\boldsymbol{g}_H}{\partial\boldsymbol{\theta}^T}. \quad (59)$$

The entries of $\boldsymbol{G}_H$ follow from (52). For unbiased channel estimates $\widehat{\boldsymbol{H}}$, let $\widehat{\boldsymbol{g}}_H$ denote the corresponding real-valued channel vector. Applying the CRB for vector parameter transformations [32, Sec. 3.8, Eq. (3.30)] gives

$$\mathrm{Cov}(\widehat{\boldsymbol{g}}_H) \succeq \boldsymbol{G}_H\overline{\boldsymbol{J}}_\theta^{-1}\boldsymbol{G}_H^T.$$

The right-hand side incorporates the effects of gain and Doppler estimation errors and their correlations on channel estimation. Under unbiased estimation, the channel mean squared error equals the trace of this covariance:

$$\mathbb{E}\Big[\|\widehat{\boldsymbol{H}} - \boldsymbol{H}\|_F^2\Big] = \mathrm{tr}(\mathrm{Cov}(\widehat{\boldsymbol{g}}_H)).$$

Taking the trace of the covariance bound and dividing by the true channel energy $\|\boldsymbol{H}\|_F^2$ therefore yields the per-realization NMSE bound

$$\mathrm{NMSE}_{\mathrm{CRB}} = \frac{\mathrm{tr}\Big(\boldsymbol{G}_H\overline{\boldsymbol{J}}_\theta^{-1}\boldsymbol{G}_H^T\Big)}{\|\boldsymbol{H}\|_F^2}. \quad (60)$$

VI. Simulation Results

A. Simulation Setup

We evaluate the channel-estimation performance of BG-SBL using the NMSE of the reconstructed equivalent channel matrix. For R channel realizations, the reported value is

$$\mathrm{NMSE}_{\mathrm{dB}} = 10\log_{10}\left[\frac{1}{R}\sum_{r=1}^{R}\frac{\|\widehat{\boldsymbol{H}}^{(r)} - \boldsymbol{H}^{(r)}\|_F^2}{\|\boldsymbol{H}^{(r)}\|_F^2}\right], \quad (61)$$

where $\boldsymbol{H}^{(r)}$ and $\widehat{\boldsymbol{H}}^{(r)}$ denote the true and estimated channel matrices. The errors are averaged in the linear domain before conversion to dB. The detected-path-count and channel-NMSE results are averaged over 100,000 independent realizations per operating point, with common channel and noise realizations used across estimators and channel realizations reused across SNR values.

We consider QPSK-modulated AFDM with $N = 256$, $k_{\max} = 4$, and $N_v = 1$. The chirp parameter c_1 follows (4), and $c_2 = \pi$. Integer delays are sampled uniformly without replacement from $\{0, \ldots, \ell_{\max}\}$. The continuous normalized Dopplers follow $\nu_p = k_{\max} \cos \theta_p$, where $\theta_p \sim \mathcal{U}[-\pi, \pi)$. For the detected-path-count and channel-NMSE experiments, the gains are independently drawn from $h_p \sim \mathcal{CN}(0, 1/P)$, giving unit average total path power. The initial dictionary uses delay and Doppler grid spacings $r_\tau = r_\nu = 1$, with integer-delay candidates from 0 to $\ell_{\max}$ and Doppler candidates from -5 to 5.

The embedded pilot block contains $N_p = 5$ equal-energy QPSK symbols, and the data-symbol power is $P_d = 1$. The aggregate pilot-to-data energy ratio is $\rho_{\mathrm{p}} = 10 \log_{10}(E_{\mathrm{pil}}/P_d)$, where E_{pil} is the total pilot-block energy. The data-symbol SNR is $10 \log_{10}(P_d/\sigma_n^2)$, where σ_n^2 is the complex thermal-noise variance. Observations are extracted from the complete pilot-and-data frame and include residual pilot–data interference.

The baselines are LMMSE [7], OG-SBL [21], GE-SBL and GR-SBL [23], Oracle-LS, and EPA-DR [11]. Dictionary-based estimators use the same pilot observations and initial grid. LMMSE assigns a prior variance of $1/K$ per coefficient. Oracle-LS estimates only the gains using the true delays and Dopplers. EPA-DR uses a single pilot with the same total pilot energy and a detection threshold of $6\sigma_n^2$.

For the SBL baselines, the noise-prior parameters are $c = d = 10^{-6}$. The gain-variance hyperparameter is $\rho = 10^{-2}$ for GE-SBL and GR-SBL and $\rho = 0.02$ for OG-SBL. The gain-variance threshold for retaining candidates is 10^{-3}, the stopping threshold is 10^{-4}, and the maximum number of iterations is 100. GR-SBL uses the local search step $\Delta = 0.01$.

For BG-SBL, $a_\pi = b_\pi = 1.01$, $c = d = 10^{-6}$, and $\pi^{(0)} = 0.2$. The stopping threshold is $\epsilon = 10^{-4}$, and the maximum number of iterations is $T_{\max} = 80$. The inner Newton search uses at most 12 iterations and a stopping threshold of 10^{-4}.

B. Results and Analysis

Fig. 5 compares the estimated path counts versus SNR.

BG-SBL stays close to the true number of paths over the plotted SNR range. By contrast, GE-SBL and OG-SBL retain substantially more components, especially at low SNR. Conventional SBL can retain multiple highly correlated dictionary atoms representing the same physical path, leading to path-count overestimation, particularly at low SNR. GR-SBL reduces this overestimation as SNR increases, but its mean path count remains above the true value. BG-SBL combines the Bernoulli activation prior π with conditional support decisions to suppress such redundant path representations.

In a continuous-Doppler channel, neighboring atoms on a fixed or refined grid may represent the same physical path, while coefficient-wise variance learning can retain several such atoms. BG-SBL adds a path only when it provides sufficient conditional evidence. Once one of two highly correlated atoms is active, the additional evidence supplied by the other is reduced. BG-SBL suppresses redundant components through this mechanism, bringing the estimated number of paths close

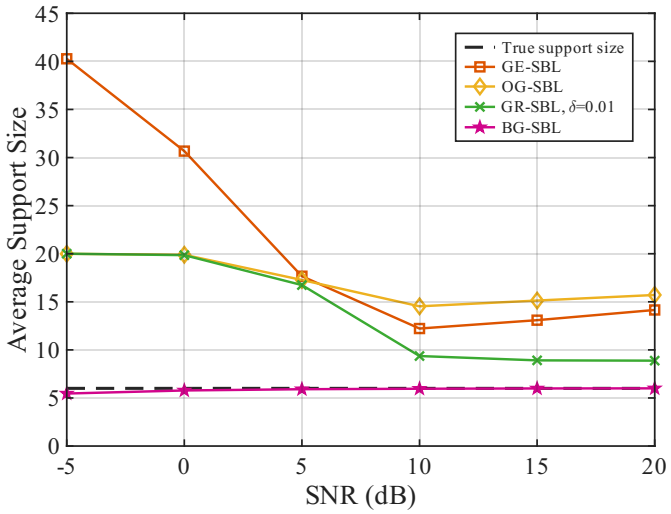


Fig. 5. Mean number of detected paths of different estimators versus SNR for $(P, \ell_{\max}) = (6, 7)$ with $\rho_{\mathrm{p}} = 30$ dB. The dashed line denotes the true number of paths.

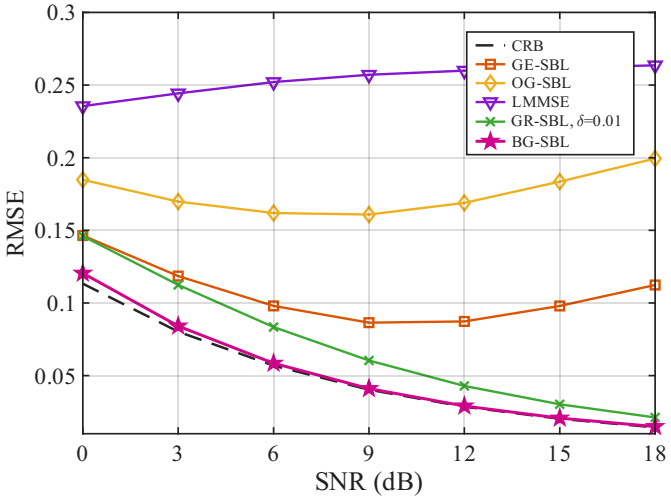


Fig. 6. Top-P success-conditioned normalized-Doppler RMSE versus SNR for $(P, \ell_{\max}) = (6, 7)$ with $\rho_{\mathrm{p}} = 20$ dB.

to the true value. However, at low signal-to-noise ratios, the gain from weak paths is insufficient to justify the cost associated with sparsity, leading to a slight underestimation of the number of paths.

The continuous-Doppler estimation results in Fig. 6 use 10,000 independent realizations per SNR. This experiment uses equal-magnitude gains with independent random phases, $h_p = e^{\mathrm{j}\varphi_p}/\sqrt{P}$, where $\varphi_p \sim \mathcal{U}[0, 2\pi)$; the delay and Doppler distributions remain unchanged.

Following the parameter-association procedure in [29], we select the P strongest estimated candidates and pair them one to one with the true paths by greedily minimizing their delay–Doppler distance. Candidates are ranked by their learned variances for the SBL methods and by coefficient energy for LMMSE. The true P is used only for evaluation. A realization is counted as a Top-P success when all selected candidates have positive strength, every paired integer-delay error is at most 10^{-8}, and every normalized-Doppler error

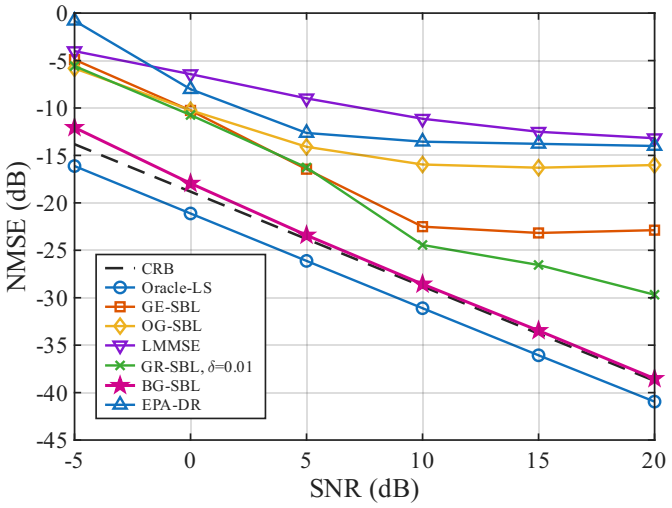


Fig. 7. NMSE performance of different channel estimators versus SNR for $(P, \ell_{\max}) = (6, 7)$ with $\rho_{\mathrm{p}} = 30$ dB.

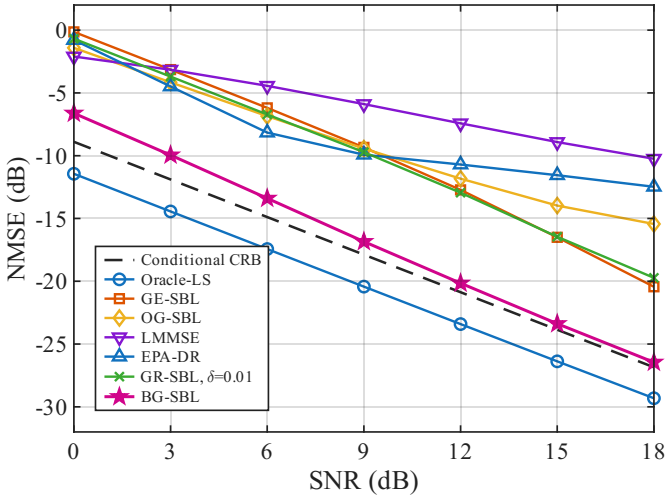


Fig. 8. NMSE performance of different channel estimators versus SNR for $(P, \ell_{\max}) = (6, 7)$ with $\rho_{\mathrm{p}} = 20$ dB.

satisfies $|\widehat{\nu}_p - \nu_p| \le r_\nu/2 = 0.5$.

Each method's RMSE is computed over its own successful realizations at a given SNR, denoted by $\mathcal{S}_m$ for method m. Let $\widehat{\nu}_{m,p}^{(t)}$ denote the estimate paired with true path p in realization t. The conditional RMSE is

$$\mathrm{RMSE}_{\nu,m} = \sqrt{\frac{1}{P|\mathcal{S}_m|} \sum_{t\in\mathcal{S}_m} \sum_{p=1}^{P} \left(\widehat{\nu}_{m,p}^{(t)} - \nu_p^{(t)}\right)^2}. \qquad (62)$$

Fig. 6 shows that BG-SBL achieves the lowest conditional Doppler RMSE across the tested SNR range, with its curve approaching the known-support CRB as SNR increases. GR-SBL also improves with SNR but remains less accurate than BG-SBL, while GE-SBL exhibits an error floor at high SNR. LMMSE and OG-SBL have larger errors. These results show that BG-SBL's continuous-Doppler refinement improves the parameter accuracy of the selected paths.

Fig. 7 compares channel-estimation NMSE versus SNR.

LMMSE has the largest error in the moderate- and high-SNR regions. OG-SBL improves on LMMSE by estimating off-grid offsets through a first-order model, but its error levels off as SNR increases. GE-SBL and GR-SBL obtain lower NMSE by updating the Doppler grid. BG-SBL achieves the lowest NMSE among the practical estimators and remains close to the conditional CRB across the plotted high-SNR region.

At high SNR, thermal noise becomes less significant, making residual dictionary mismatch a major source of estimation error. BG-SBL reduces this mismatch by evaluating exact AFDM atoms at refined continuous Dopplers, while conditional support decisions reject correlated candidates that duplicate already represented paths. Accurate path-count and Doppler estimates improve channel reconstruction and reduce NMSE.

Fig. 8 examines channel-estimation NMSE with reduced pilot energy.

The NMSE of BG-SBL continues to decrease with SNR and stays close to the conditional CRB, while LMMSE retains the largest error. GE-SBL, OG-SBL, and GR-SBL achieve lower NMSE than LMMSE but remain above BG-SBL at all tested SNRs. Thus, BG-SBL retains its NMSE advantage with reduced pilot energy and more prominent residual data interference relative to the pilot response.

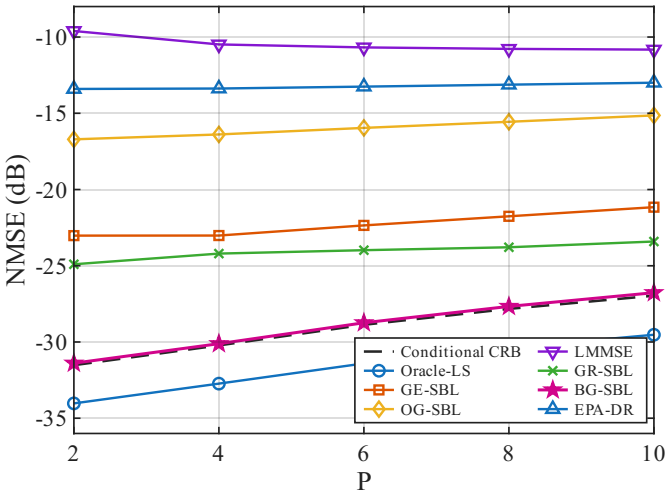


Fig. 9. NMSE performance of different channel estimators versus the number of paths P, with $\ell_{\max} = 9$, $\rho_{\mathrm{p}} = 30$ dB, and SNR $= 10$ dB.

Fig. 9 shows how channel-estimation NMSE changes with the number of paths. The pilot layout and initial dictionary are held fixed throughout this sweep.

BG-SBL achieves the lowest NMSE among the practical estimators over the tested range of path numbers and remains close to the known-support CRB. Its NMSE increases as the number of paths grows, following the trend of the CRB. GR-SBL is the closest practical baseline, and its performance gap relative to BG-SBL narrows as P increases. With the observation size and average total path power fixed, more paths introduce additional unknown gains and Dopplers while reducing the average power per path, which is consistent with the observed increase in estimation error.

VII. Conclusion

This paper investigates the problem of off-grid channel estimation for AFDM systems under fractional Doppler conditions. This paper proposes a Bernoulli–Gaussian sparse Bayesian learning (BG-SBL) method to address dictionary mismatch under fractional Doppler and path-count overestimation due to redundant path representations by highly correlated candidate atoms. This method explicitly learns the path support while jointly updating the continuous Doppler coordinates and channel gains of active paths. Simulation results demonstrate that, for the tested channel configuration, BG-SBL maintains the estimated number of paths close to the true value and achieves a superior channel estimation Normalized Mean Square Error (NMSE) compared to benchmark methods; this advantage persists even when pilot energy is reduced. For samples satisfying the path matching condition, the Doppler estimation error approaches the Cramér-Rao Bound (CRB) reference curve for known support. These results indicate that the combination of explicit path support learning and continuous Doppler refinement effectively enhances off-grid channel estimation performance in AFDM systems.

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